Onyx — PrinterKit reference

Onyx · multi-process OS on Raspberry Pi 4

*The reference of **PrinterKit** (`SD:/lib/printerkit.so`, `user/Kits/printerkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: The Kits.*

Contents

1. What it is
2. Using it
3. Index
4. `printerkit/printerkit.h`

What it is

PrinterKit is printing: the Print dialog, a job and its pages — text, lines, shapes, pictures — handed to the print service, which makes of them what the printer takes, or a PDF file.

Include	`#include "printerkit/printerkit.h"`
Link	`lib/printerkit.imp.a`
Library	`SD:/lib/printerkit.so` — 33 entries in its table (`user/Kits/printerkit/printerkit.abi`, append-only)
Sources	`user/Kits/printerkit/`

Using it

A program prints in four steps: the Print dialog, a job, its pages, the end. Lengths are points (1/72 inch) from the page's top-left corner; colours are `0xRRGGBB`. The pages are recorded: the print service turns them into what the printer takes — or into a PDF file.

```
#include "printerkit/printerkit.h"

PrintSetup s;
print_setup_default (&s);
PrintDialogInfo di = { sizeof di, "My document", 1, 0, 0, 0, 0 };
if (print_dialog (&s, &di))                          // the printer, the paper, the pages
{
    PrintJob *j = print_begin (&s, "My document");
    int font = print_font (j, "DejaVu Sans", 0);
    if (print_page (j, 0, 0))                        // 0, 0: the paper chosen
    {
        print_text (j, font, 12, 72, 84, "Hello, printer", 0x000000);
        print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
    }
```

```
    print_end (j);                                  // to the print queue
}
```

Rectangles, paths (lines and curves, filled or stroked) and pictures are drawn the same way (`print_rect`, `print_path_*`, `print_image`).

Index

Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
`PrintSetup`	What the user chose in the Print dialog.	`printerkit.h`
`PrintDialogInfo`	the page is fitted on the printer's paper	`printerkit.h`
`print_setup_default`	the default printer, its default paper, upright, one copy	`printerkit.h`
`print_dialog`	1: print (s filled), 0: cancelled	`printerkit.h`
`print_setup_paper`	another paper → 1, 0 unknown name	`printerkit.h`
`print_begin`	0: the spool cannot be written	`printerkit.h`
`print_page`	a new page (w, h 0: the setup's paper) → 1, or 0: outside	`printerkit.h`
`print_rect`	the pages to print -- skip its drawing (it would be ignored)	`printerkit.h`
`print_line`	the pages to print -- skip its drawing (it would be ignored)	`printerkit.h`
`print_frame`	a rectangle's outline	`printerkit.h`
`print_path_move`	a shape	`printerkit.h`
`print_path_line`	a shape	`printerkit.h`
`print_path_curve`	a shape	`printerkit.h`
`print_path_close`	a shape	`printerkit.h`
`print_path_fill`	a shape	`printerkit.h`
`print_path_stroke`	a shape	`printerkit.h`
`print_image`	an image	`printerkit.h`
`print_font`	text	`printerkit.h`
`print_text`	text	`printerkit.h`
`print_text_width`	text	`printerkit.h`
`print_font_metrics`	text	`printerkit.h`
`print_font_data`	for an app with its own	`printerkit.h`

Name	What it does	Header
	text layout	
`print_glyph`	for an app with its own text layout	`printerkit.h`
`print_end`	the job queued → its number (> 0), or < 0: it could not be	`printerkit.h`
`print_abort`	nothing printed	`printerkit.h`
`PrintPrinter`	(a type)	`printerkit.h`
`print_printers`	→ how many	`printerkit.h`
`print_printer_media`	a paper of a printer	`printerkit.h`
`print_printer_add`	1 / 0 (err: why)	`printerkit.h`
`print_printers_find`	the printers found on the network into out (max, 16 at most) → how many (0: none, no print service)	`printerkit.h`
`print_printer_remove`	taken off the list, by its name → 1 / 0 (not known; the PDF printer stays)	`printerkit.h`
`print_printer_default`	made the default printer, by its name → 1 / 0 (not known)	`printerkit.h`
`print_printer_status`	asked now: 1 ready (text: ink...), 0 (text: why)	`printerkit.h`
`print_jobs`	→ how many (the queue, then the last finished)	`printerkit.h`
`print_job_cancel`	a waiting or printing job stopped → 1 / 0 (no such job, no print service)	`printerkit.h`
`print_jobs_forget`	the finished jobs dropped from the list	`printerkit.h`

`printerkit/printerkit.h`

printerkit/printerkit.h -- printing from an app (the shared library SD:/lib/printerkit.so; docs/03 "Printing").

An app prints in four steps -- the Print dialog, a job, its pages, the end:

```
  PrintSetup s; print_setup_default (&s);
  PrintDialogInfo di = { sizeof di, "My document", pageCount, 0, 0, 0, 0 };
  if (print_dialog (&s, &di))                          // the user chose a printer,
a paper, the pages...
```

```
  {
      PrintJob *j = print_begin (&s, "My document");
      int f = print_font (j, "DejaVu Sans", 0);
      for (int p = 0; p < pageCount; p++)
          if (print_page (j, 0, 0))                     // 0, 0: the paper chosen; 0
returned: not in the range
          {
              print_text (j, f, 12, 72, 72 + 12, "Hello", 0x000000);
              print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
          }
      print_end (j);                                    // the job goes to printd's
queue
  }
```

Lengths are points (1/72 inch), y goes down from the page's top-left corner, colours are 0xRRGGBB. The pages are recorded, not rendered: the print service (printd) makes of them what the printer takes -- a PDF file for the PDF printer, pages of pixels at its resolution for a network printer -- and tells the user when the job is done. The app does not wait for the printer.

A program links lib/printerkit.imp.a (with lib/uikit.imp.a and lib/fontkit.imp.a: the dialog is uikit's, the text FreeType's). The table is append-only (printerkit/printerkit.abi): these functions never change.

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```
typedef struct PrintJob PrintJob;

enum { PRINT_DRAFT = 3, PRINT_NORMAL = 4, PRINT_HIGH = 5 };     // quality
enum { PRINT_PORTRAIT = 0, PRINT_LANDSCAPE = 1 };
enum { PRINT_BOLD = 1, PRINT_ITALIC = 2 };              // a font's style
enum { PRINT_IMG_ALPHA = 1,     // the pixels are 0xAARRGGBB (else 0x00RRGGBB:
opaque)
       PRINT_IMG_PHOTO = 2 };       // a photo: kept as a JPEG (smaller; not for text
or line art)
```

What the user chose in the Print dialog. Fields are only ever added at the end (size says how many).

```
typedef struct PrintSetup
{
    unsigned size;              // sizeof (PrintSetup)
    char  printer[64];          // the printer's name ("PDF": the PDF printer)
    char  media[64];            // the paper, as IPP names it ("iso_a4_210x297mm")
    float paper_w, paper_h;         // the page in points, the orientation applied
    float margin_l, margin_t, margin_r, margin_b;   // what the printer cannot print
on, along each edge
    int   orientation;          // PRINT_PORTRAIT / PRINT_LANDSCAPE
    int   copies;
    int   mono;             // 1: black and white
    int   quality;              // PRINT_DRAFT / NORMAL / HIGH
    int   from, to;             // the pages to print, from 1; 0, 0: all
    char  output[256];          // the PDF printer: the file to write
```

```
    int   reserved[16];
} PrintSetup;

enum { PRINT_DLG_OWN_PAPER = 1 };   // the document has its own page size (a slide, a
PDF): no paper choice --
```

the page is fitted on the printer's paper

```
typedef struct PrintDialogInfo
{
    unsigned size;              // sizeof (PrintDialogInfo)
    const char *title;          // the document's name (the dialog's title, the PDF's
file name)
    int   pages;                // how many pages it has (0: not known)
    int   current;              // the page shown (from 1: offers "Current page"; 0:
no such choice)
    int   flags;                // PRINT_DLG_*
    float paper_w, paper_h;          // the document's own paper to select first (0:
the printer's default)
} PrintDialogInfo;
```

the dialog

```
void print_setup_default (PrintSetup *s);           // the default printer, its
default paper, upright, one copy
int  print_dialog (PrintSetup *s, const PrintDialogInfo *info); // 1: print (s
filled), 0: cancelled
int  print_setup_paper (PrintSetup *s, const char *media, int orientation); //
another paper -> 1, 0 unknown name
```

a job

```
PrintJob *print_begin (const PrintSetup *s, const char *title); // 0: the spool
cannot be written
int   print_page (PrintJob *j, float w, float h);   // a new page (w, h 0: the
setup's paper) -> 1, or 0: outside
```

the pages to print -- skip its drawing (it would be ignored)

```
void  print_rect (PrintJob *j, float x, float y, float w, float h, unsigned rgb);
void  print_line (PrintJob *j, float x0, float y0, float x1, float y1, float width,
unsigned rgb);
void  print_frame (PrintJob *j, float x, float y, float w, float h, float width,
unsigned rgb); // a rectangle's outline
```

a shape: move to, lines and Bezier curves, closed or not; then filled, or stroked with a line

```
void  print_path_move (PrintJob *j, float x, float y);
void  print_path_line (PrintJob *j, float x, float y);
void  print_path_curve (PrintJob *j, float x1, float y1, float x2, float y2, float
x3, float y3);
void  print_path_close (PrintJob *j);
void  print_path_fill (PrintJob *j, unsigned rgb);
void  print_path_stroke (PrintJob *j, float width, unsigned rgb);
```

an image: pw x ph pixels drawn in (x, y, w, h)

```
void  print_image (PrintJob *j, const unsigned *px, int pw, int ph, float x, float y,
float w, float h, int flags);
```

text: a font by its family ("DejaVu Sans", "Liberation Serif"...: SD:/res/fonts, SD:/fonts) and style → its number (-1: no such font nor a fallback); the text at its baseline → the pen's advance

```
int   print_font (PrintJob *j, const char *family, int style);
float print_text (PrintJob *j, int font, float size, float x, float baseline, const
char *utf8, unsigned rgb);
float print_text_width (PrintJob *j, int font, float size, const char *utf8);
void  print_font_metrics (PrintJob *j, int font, float size, float *ascent, float
*descent, float *line);
```

for an app with its own text layout: a TrueType file's bytes (kept by the app until print_end) → the font's number; one glyph of it at its pen position (style: PRINT_BOLD / ITALIC made from this face)

```
int   print_font_data (PrintJob *j, const unsigned char *ttf, unsigned len);
void  print_glyph (PrintJob *j, int font, float size, float x, float baseline,
unsigned glyph, unsigned unicode, unsigned rgb, int style);
int   print_end (PrintJob *j);      // the job queued -> its number (> 0), or < 0: it
could not be
void  print_abort (PrintJob *j);    // nothing printed
```

the printers and the queue (the Control Panel's applet; an app's own status)

```
typedef struct PrintPrinter
{
    char name[64], kind[8], uri[160], model[64];    // kind: "pdf", "ipp"
    int  color, copies, dpi, is_default;
    unsigned quality;               // bits 3 (draft), 4 (normal), 5 (high)
    char media_default[64];
} PrintPrinter;
enum { PRINT_JOB_QUEUED = 0, PRINT_JOB_PREPARING, PRINT_JOB_SENDING,
PRINT_JOB_PRINTING, PRINT_JOB_DONE, PRINT_JOB_FAILED, PRINT_JOB_CANCELED };
typedef struct PrintJobInfo { unsigned id; int state, pages, page; char printer[64],
title[80], message[96]; } PrintJobInfo;

int  print_printers (PrintPrinter *out, int max);       // -> how many
```

a paper of a printer: its IPP name, its name for a person → 1, 0: no i-th paper

```
int  print_printer_media (const char *printer, int i, char *name, int ncap, char
*label, int lcap);
```

a network printer asked what it can do, and added (address: "192.168.0.14", "ipp://host:631/ipp/print")

```
int  print_printer_add (const char *name, const char *address, char *err, int cap);
// 1 / 0 (err: why)
```

the local network searched for printers (mDNS / DNS-SD, as AirPrint: a few seconds) → how many; usable 0: it answers but takes neither PWG Raster nor PDF

```
typedef struct PrintFound { char address[32], name[64], model[64]; int usable; }
PrintFound;
int  print_printers_find (PrintFound *out, int max);    // the printers found on the
network into out (max, 16 at most) -> how many (0: none, no print service)
int  print_printer_remove (const char *name);   // taken off the list, by its name ->
1 / 0 (not known; the PDF printer stays)
int  print_printer_default (const char *name);  // made the default printer, by its
name -> 1 / 0 (not known)
int  print_printer_status (const char *name, char *text, int cap);  // asked now: 1
ready (text: ink...), 0 (text: why)
int  print_jobs (PrintJobInfo *out, int max);           // -> how many (the queue,
then the last finished)
int  print_job_cancel (unsigned id);    // a waiting or printing job stopped -> 1 / 0
(no such job, no print service)
```

```
void print_jobs_forget (void);                  // the finished jobs dropped from the
list

}
#endif
```